

1. Standard Workflow (Steps to Follow Each Time)

A. Prepare Sentry for Bug Testing

Step 1: Create the right Sentry projects

- â€¢ Create a frontend project for the browser app.
- â€¢ Create a backend project if your server talks to the database.
- â€¢ Copy the DSN for each project from Sentry.

Step 2: Add environment variables

- â€¢ Store the frontend and backend DSNs in environment files.
- â€¢ Set environment values like staging or production.
- â€¢ Set a release name so issues can be tied to deployments.

```
VITE_SENTRY_DSN=https://public@example.ingest.sentry.io/123456
SENTRY_DSN=https://public@example.ingest.sentry.io/654321
SENTRY_ENVIRONMENT=staging
SENTRY_RELEASE=webapp-1.0.0
```

Step 3: Install the SDKs

```
npm install @sentry/react @sentry/browser
npm install @sentry/node
```

Use the frontend SDK for browser-side issues and the backend SDK for API and database-related errors.

Step 4: Initialize Sentry in the application

- â€¢ Initialize Sentry at frontend startup.
- â€¢ Initialize Sentry as early as possible in the backend.
- â€¢ Restart the app after adding the DSN and config.

```
Sentry.init({
  dsn: import.meta.env.VITE_SENTRY_DSN,
  environment: import.meta.env.MODE,
  release: "webapp-1.0.0",
  tracesSampleRate: 1.0,
});
```

B. Run Bug Tests / Common Testing Patterns

1. Test a handled frontend exception with `captureException`.

2. Test an unhandled frontend crash with `throw new Error(...)`.
3. Test an unhandled promise rejection.
4. Test a backend route that throws a controlled error.
5. Test a database/query failure through the backend.
6. Test a failed API flow seen by both frontend and backend.

2. Detailed Test Procedure

A. Frontend Bug Tests

Step 1: Verify the SDK is active

- â€¢ Open the web app in a browser.
- â€¢ Open DevTools and keep the Network tab visible.
- â€¢ Look for requests to `sentry.io`, `ingest.sentry.io`, or `/envelope/` after a test event.

Step 2: Trigger a handled exception

```
Sentry.captureException(new Error("Frontend handled test bug"));
```

- â€¢ Confirm that an issue appears in Sentry.
- â€¢ Check the stack trace, browser, URL, environment, and release.

Step 3: Trigger an unhandled crash

```
throw new Error("Frontend unhandled crash test");
```

- â€¢ Confirm the event is captured separately from the handled test.
- â€¢ If an error boundary exists, confirm the fallback UI appears.

Step 4: Trigger an unhandled promise rejection

```
Promise.reject(new Error("Unhandled promise rejection test"));
```

B. Backend and Database Bug Tests

Step 5: Trigger a backend exception

```
app.get("/test-backend-error", (req, res) => {  
  throw new Error("Backend crash test");  
});
```

Step 6: Trigger a database error through the backend

```
app.get("/test-db-error", async (req, res, next) => {
  try {
    const result = await db.query("SELECT * FROM missing_table");
    res.json(result.rows);
  } catch (error) {
    Sentry.captureException(error);
    next(error);
  }
});
```

Sentry does not test the database directly. It captures the database error when your backend throws or records it.

- â€¢ Use safe staging-only failures like a missing table, missing column, timeout, or duplicate key test data.
- â€¢ Do not use destructive production tests.

3. Verification and Cleanup

A. What to Verify in Sentry

1. Issue title is clear and grouped correctly.
2. Stack trace points to the right source files.
3. Environment and release values are present.
4. User, tags, and breadcrumbs are visible if configured.
5. Frontend and backend failures can be correlated when the API fails.

B. Common Safe Context to Add

```
Sentry.setUser({ id: "123", email: "tester@example.com" });
Sentry.setTag("area", "checkout");
Sentry.setContext("test_case", { name: "db-timeout-simulation" });
```

C. Final Checklist

1. DSN present and SDK installed.
2. Frontend and backend initialized.
3. Handled and unhandled frontend issues captured.
4. Backend and DB-related issues captured.
5. Source maps, grouping, and alerts verified.
6. Temporary crash buttons and test routes removed.